

Breast Cancer Incidence and Mortality Trends in Missouri

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Abstract

Among women in Missouri as in the United States (U.S.), breast cancer is the leading site of new cancer cases and the second leading cause of cancer deaths. This study examined patterns of breast cancer incidence and mortality among Missouri women. Age-adjusted Missouri incidence (1985-1992) and mortality (1985-1994) data were compared with Surveillance, Epidemiology, and End Results (SEER) data (1985-1992) by race and diagnosis year. Missouri Cancer Registry (MCR) data indicate that there have been no significant changes in breast cancer incidence and mortality rates through the time periods. Incidence rates for Missouri females (White, African-American and All) are significantly lower than SEER rates for all age groups, with greater differences in all races among women over 65. Incidence rates for African-Americans are significantly lower than for whites. However, Missouri mortality rates for African-Americans fluctuate over the period, but are consistently higher than for whites. In addition, African-Americans over age 65 have higher mortality rates in Missouri than in the U.S.

Conclusions

Patterns and trends of incidence and mortality are compatible with increased mammography screening. However, results indicate that screening efforts must be increased in Missouri.

Introduction

Breast cancer represents the most common invasive malignant condition affecting women from developed countries, with the highest rates of inci-

dence observed in white U.S. females¹ followed by women living in northern Europe. Intermediate incidence rates are reported in southern Europe and South America, while the lowest rates occur in Asia and Africa.² While a sharp rise in the incidence of breast cancer characterized the 1980s, mostly reflecting increased use of mammography, breast cancer mortality rates have declined, particularly in white women.³ Recent studies indicate that five-year survival rates of African-American women still trail behind those of white women, with rates of 64% and 80%, respectively.⁴

Research from the past 20 years has stressed several important breast cancer risk factors. Various reproductive factors are known to be tied to increased rates of breast cancer incidence. Of these, nulliparity, early age at menarche, late age at first full term birth, and late age at menopause are most commonly cited.^{5,3} The common denominator behind these risk factors seems to be reproductive hormone levels.⁵

Family history and genetics also play a role.² For instance, breast cancer risk increases substantially for women whose mother and sister have both had breast cancer at young ages.²

A recent study looks at the possible effects of diet on breast cancer rates⁶ by examining adult intake of soy among breast cancer cases from various Asian-American groups from Los Angeles County Metropolitan Statistical Area, San Francisco Oakland Metropolitan Statistical Area, and Oahu, Hawaii. Results indicate a decrease in breast cancer risk with increased frequency of soy intake. Another study suggests that dietary fat is tied to a higher incidence of breast cancer.²

Exercise may also play a role as a risk factor for breast cancer. Low levels of physical activity have been tied to increased risks of breast cancer.^{7,8}

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While breast cancer incidence varies between races,³ several recent studies suggest that risk factors for African-American women are similar to those for white U.S. women.^{9,4} However, African-American women have significantly higher breast cancer mortality rates than U.S. white women.^{9,3,4} One study⁴ suggests that one reason for this disparity may be a difference in tumor biology, specifically that African-American women have higher frequencies of poorly differentiated tumors and lower frequencies of hormone receptor-positive tumors. A study by Schatzkin et al.⁹ suggests a link between obesity and breast cancer in postmenopausal African-American women. However, socioeconomic factors and lower rates of early detection continue to be the important reasons behind the fact that African-American women die of breast cancer at higher rates than white women⁴ in the U.S.

Among women in Missouri as in the U.S., the breast is the leading site of new cancer cases and the second leading cause of cancer deaths. This report reviews patterns and trends of breast cancer incidence and mortality in Missouri between 1985 and 1992 and 1985-1994, respectively, with particular attention to differences due to age and race. Comparable national figures are taken from the National Cancer Institute's (NCI) SEER program.¹⁰ Because breast cancer is rarely diagnosed among men (less than 1%), this report focuses solely on women.

Methods

Using incidence data collected between 1985 and 1992 by the Missouri Cancer Registry (MCR), and mortality data collected between 1985 and 1994 by the Center for Health Information Management and Epidemiology (CHIME), we calculated incidence and mortality rates with corresponding 95% confidence intervals by year, age, gender, and race (White, African-American, and All). We used CHIME estimates of annual Missouri population size based on intercensus extrapolations for rate calculations. In addition, to control for effects of a shifting age distribution of the population, we used direct methods and the 1970 U.S. standard million population to age-adjust rates. We assessed temporal trends in incidence and mortality of breast cancer by generating annual and period percent change in rates (1985-1992 and 1985-1994) for incidence and mortality, respectively. A period percent change was calculated by taking the difference between the average of the two halves of the studied period rates divided by the average of the first half period rates and multiplying by 100 as follows:

$$\text{Percent Change in Incidence Rate} = \frac{\text{Average of 2nd half} - \text{Average of 1st half}}{\text{Average of 1st half}} * 100$$

and

$$\text{Percent Change in Mortality Rate} = \frac{\text{Average of 2nd half} - \text{Average of 1st half}}{\text{Average of 1st half}} * 100$$

We divided the period percent change by the number of years in the rate series (eight in the incidence rate and 10 in the mortality rate) to estimate the average annual percent change in rates. We determined the statistical significance of rate differences by setting 95% confidence intervals around the rate for each year.

Results

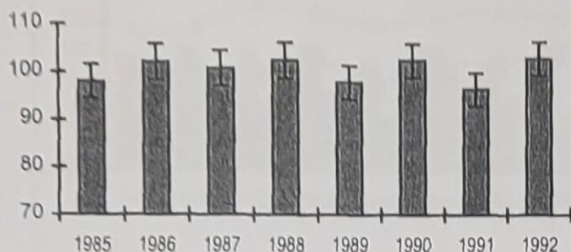
1. Incidence

Missouri incidence rates of breast cancer for all races fluctuated between 1985 and 1992, with no significant change (Figure 1A). During that period, incidence rates for both white and African-American women decreased slightly (4% and 2.5%, respectively), but none of the changes were statistically significant (Figures 1B, 1C, and 3A). Average incidence rates for years 1985-1988 and 1989-1992 for white Missouri women were 101.1 and 100.7 per 100,000. For African-American women, average rates for 1985-1988 and 1989-1992 were 86.8 and 84.6 per 100,000. However, between 1985 and 1992, the incidence of localized and in-situ stages increased 17.5% and 56%, respectively. Regional and distant stages decreased 16% and 22%, respectively, during the same period (not shown in figures).

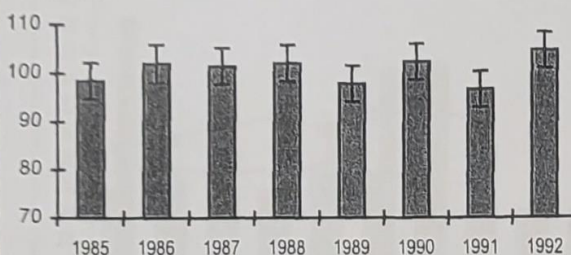
For all ages combined, the Missouri incidence rate of breast cancer

Figure 1. Missouri Yearly Incidence Rates with 95% Confidence Intervals of Breast Cancer for Women only for (A) All Races, (B) Whites, and (C) African-Americans.

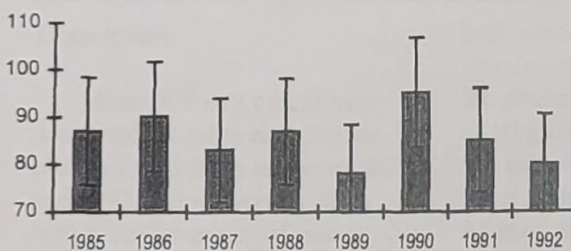
A. All Races



B. Whites



C. African-American



was 15.8% higher among white women compared to that of African-American women throughout the 1985-1992 period. This racial difference increased to 21.5% among women ages 65 and older.

Among African-American women, those age 65 and older had an incidence rate five times higher than that of those younger than 65. Likewise, the incidence rate for white women 65 and above was six times higher than that of younger women.

2. Mortality

Overall, the Missouri breast cancer mortality rate for all women declined by 4.9% between 1985 and 1994 (Figure 2A). This trend was characterized

in both white and African-American women, with decreases of 4.8% and 3.8%, respectively, although none of the changes were statistically significant (Figures 2B, 2C and 3B). Average mortality rates for years 1985-1988 and 1989-1992 for white Missouri women were 26.2 and 24.9 per 100,000. For African-American women, average rates for 1985-1988 and 1989-1992 were 32.3 and 31.1 per 100,000.

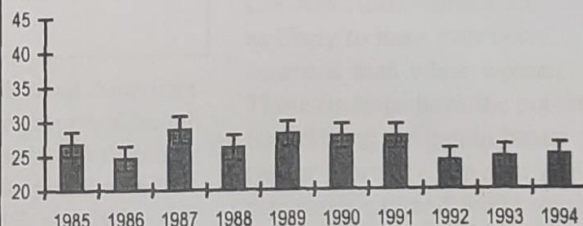
Overall mortality rates for African-American women under age 65 were 37.4% higher than those of white women from the same age group. This difference dropped to 5.4% for women age 65 and older.

Older women (ages 65 and older) had a mortality rate nearly 40 times higher than younger women. This difference was seen in both white and African-American women.

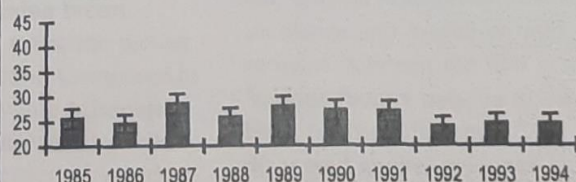
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Figure 2. Missouri Yearly Mortality Rates with 95% Confidence Intervals of Breast Cancer for Women only for (A) All Races, (B) Whites, and (C) African-Americans.

A. All Races



B. Whites



C. African-Americans

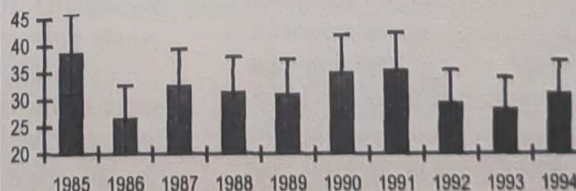
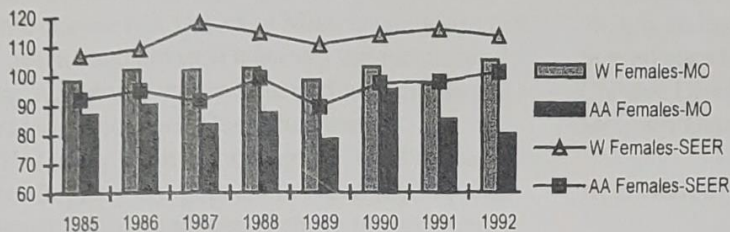
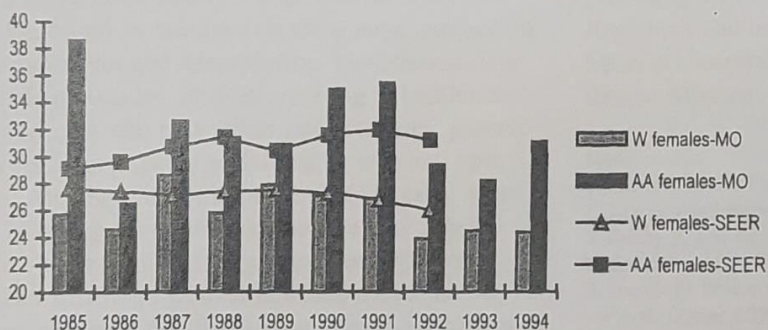


Figure 3. Missouri and National (SEER) Incidence (A) and Mortality (B) Rates of Breast Cancer, White and African-American Women.

A. Incidence



B. Mortality



Discussion

Missouri breast cancer incidence and mortality rates did not change significantly between 1985 and 1992. White women had a higher incidence rate of cancer than African-American women, particularly in the age group 65 and above. For both racial groups, incidence and mortality rates were much higher in older women. However, younger African-American women from Missouri were much more likely to die of breast cancer than young white Missouri women.

When compared with U.S. rates during the same period, Missouri breast cancer incidence rates were lower than national estimates.¹⁰ This was true for

both white and African-American women, in all age groups, although the greatest difference for both racial groups is seen in the >65 age category (23% and 20% lower difference, for whites and African-Americans, respectively).

When examining breast cancer mortality rates, the picture looks quite different. Compared to national rates,¹⁰ white Missouri women of all ages were less likely to die of breast cancer than other U.S. white women. On the other hand, breast cancer mortality rates for Missouri African-American women were above national levels.

Because risk factors associated with breast cancer account for only a small percentage of the disease, and few of those risks

can be altered either by environmental or behavioral modification,² primary prevention of breast cancer remains difficult. So far, most studies point to regular and systematic use of mammography as the best means of detecting breast tumors at an early stage. Early detection of breast cancer tumors (still localized) can lead to five-year survival rates as high as 97%.¹¹ Estimates of reduction in breast cancer mortality resulting from mammography screening range between 17% and 33%.²

This report indicates through a shift in early stage diagnoses that early detection has substantially increased in Missouri between 1985 and 1992. A recent study of mammography utilization in Missouri between 1993-1994¹² suggests that African-American women are twice as likely to have ever been screened than white women. These findings have the potential for bridging the gap in breast cancer mortality between African-American and white women in Missouri. However, as results from this study show, a large disparity in both breast cancer incidence and mortality rates remains between the two groups. Various factors may lie at the heart of this difference. As discussed above, evidence exists for higher frequencies of poorly differentiated tumors in African-American women,⁴ which might contribute to the racial discrepancy in mortality rates due to breast cancer. It is also possible that African-American women may suffer from insufficient health care availability, making

clinical management of breast cancer among these women less effective.

Additionally, Schatzkin et al.⁹ suggest a link between obesity and breast cancer in postmenopausal African-American women. A recent study of chronic disease risk factors in Missouri indicates that both obesity and physical inactivity are increasing among African-American Missouri women, particularly among those with less education.¹³

This study has many strengths. The Missouri Cancer Registry is a population-based registry with estimated 92% completeness, above the minimum standard required by the North American Association of Central Cancer Registries (personal communication; Dr. Jian Chang). Rate estimates are reliable, based on calculations using large numbers in both numerator and denominator. Nonetheless, it is possible that under- or over-reporting of incidence for a specific site may affect rate estimates, particularly for a subpopulation defined by race and age.

Although recent improvements in early detection of malignant tumors has the potential to help narrow the mortality rate gap between white and African-American women in Missouri, the breast cancer mortality rate for African-American Missouri women remains above the national rate, indicating that screening efforts must be continued and should be increased. Recently approved legislation on health insurance coverage may substantially increase mammography availability for Missouri women. Future analyses of updated incidence data by stage, mortality data, and trends in mammography utilization should be sufficient to determine more recent trends and effects of early detection on overall breast cancer mortality in Missouri. **MoMed**

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